

Interactive Live Site Diagram

A short, self-contained take-home for frontend engineers

Thanks for taking this on. **BlackTeal** builds monitoring software for battery energy storage systems (BESS) that support large electrical loads such as data centers. This exercise mirrors real work you'd do on the team: building operator-facing interfaces for live power-system data.

Please build this as soon as you can. We're keen to see it working — prioritize a clean, functional result over exhaustive polish, with a short README covering how to run it and the trade-offs you made. We want to see how you structure code, handle live data, and deal with edge cases.

The scenario

A site has a grid connection through a **substation**, several **power skids** (each skid contains a transformer, an inverter, and a battery), and a **data-center load**. Operators watch this site live and need to spot and diagnose problems quickly.

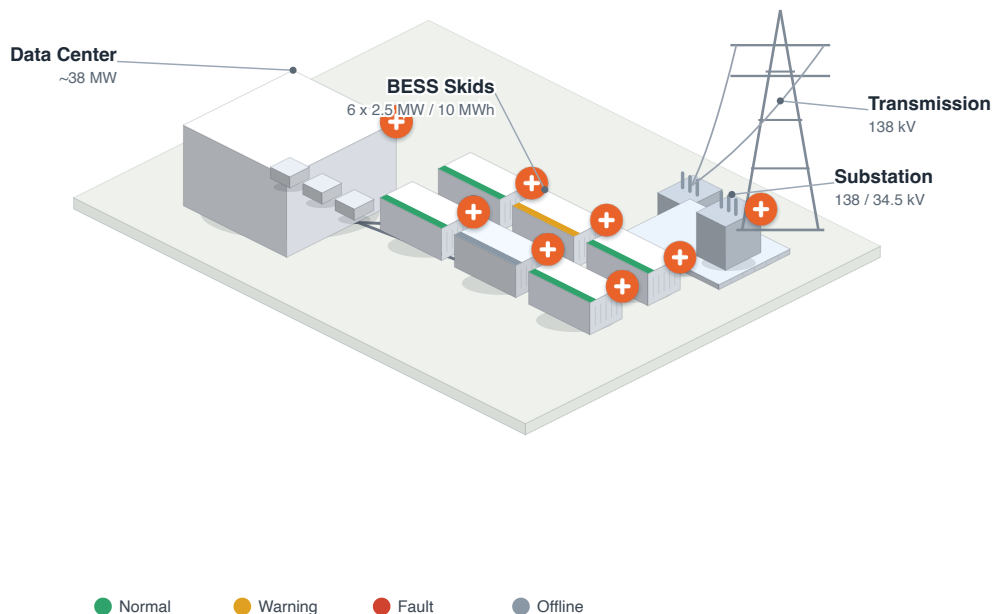


Figure 1 — Reference site layout. The grid/substation feeds six power skids (each = transformer + inverter + battery), which supply the data-center load. The “+” markers indicate assets an operator can click to inspect — that click-to-inspect interaction is what you'll build. (A simple diagram like this is perfectly fine; the art is not what we're evaluating.)

Your task

Build a small web app showing an **interactive diagram** of the site, where:

1. **Each asset shows its live status** — NORMAL / WARNING / FAULT / OFFLINE — plus a key metric.
2. **Clicking an asset opens a detail/debug panel** with its live telemetry, any active alarms, and its connection health.
3. **Data updates live**, and the UI **clearly shows when data is stale/disconnected** (never present frozen values as live).

What we provide

Everything you need is in the **Site Data Pack** at the end of this document: the site topology, nameplate specifications, the telemetry schema with units and normal ranges, an alarm catalog, and a realistic snapshot to render. No backend, accounts, or external data required — simulate the live feed yourself (guidance included).

Requirements must-have

- Render the site from the topology: the substation, the skids, and the load, with the connections between them.
- Show **live status** per asset using **color and a label/icon** (don't rely on color alone).
- Click an asset → a **detail panel** with its live telemetry, active alarms, and connection health.
- **Live updates**, plus a clear **stale/disconnected** indication when the feed stops.
- A short **README**: how to run it (ideally one or two commands), and notes on your trade-offs and what you'd do with more time.

Build these stages

Get the core (above) working first, then extend it. **Stages 1–2 are what we most want to see**; 3–4 are bonus.

Stage 1 — Alarm console expected

- A site-wide **alarm list, priority-sorted** (critical above warning), showing the asset and message.
- **Acknowledge** and **shelve** (temporarily silence) an alarm.
- **Filter** by asset and severity.
- **Flood handling**: when many related alarms arrive at once, **roll them into one grouped, counted entry** — not a wall of rows.

Why it matters: real operators can act on only ~6–12 alarms/hour, so surfacing the one that matters — and not burying it — is the whole job.

Stage 2 — Operator-grade styling expected

- Follow the **High-Performance HMI** convention: keep most of the screen **calm and neutral (gray)**, and use **color only to mean “look here now.”**

- Muted color for warnings, saturated for critical. A healthy site should look quiet; an abnormal one should jump out at a glance.

Why it matters: it's the control-room standard (ISA-101) — a busy, colorful dashboard actually slows operators down.

Stage 3 — Trends & headroom bonus

- In the detail panel, a **sparkline** of a key metric's recent history.
- Show each battery's **headroom against its operating envelope** — how much charge/discharge margin is left, and why it's derated.

Stage 4 — Explain the state bonus

- For each abnormal asset, one **plain-language line** — e.g. “Skid 2 derated: a module is at 41 °C, so discharge is capped to 1.5 MW.” Simple rules are fine (no AI needed).

Also welcome (small extras)

- Power-flow direction/magnitude on the connections; a couple of tests around your state/data logic; zoom/pan or a responsive layout.

Out of scope please don't spend time here

- **No control actions.** This exercise is monitoring/visualization only — read-only.
- No authentication, no real backend, no database.
- No pixel-perfect graphics. A clean, simple diagram (boxes, circles, lines) is exactly right — we care about the *interaction* and *data handling*, not the art.

What we're looking for

- Clear, readable, well-organized code.
- Sensible handling of real-time updates and edge cases — missing metrics, an offline asset, a stale feed, alarm bursts.
- **Alarm handling that cuts noise** — the operator sees what matters, not a wall of rows.
- **Situational awareness** — can someone glance at the screen and instantly tell whether the site is healthy?
- A drill-down an operator could actually use: after clicking an asset, can you understand its situation quickly?
- Good judgment on trade-offs, explained briefly in your README.

Tech & submission

- Use whatever you're most productive in. Our stack is **React + TypeScript**, so that's welcome, but pick what lets you do your best work.
- Deliver a **runnable app**: a link to a Git repository (or a zip), including the README.

- Send it to **careers@blackteal.com**. Questions any time at the same address.

APPENDIX

Site Data Pack

Input data for the exercise. All values are representative of a real grid-connected battery + data-center site, but are synthetic — they don't come from any live system. Feel free to vary them.

1 • Plant overview

Grid interconnection	138 kV, 3-phase, 60 Hz
Battery energy storage (BESS)	15 MW / 60 MWh (4-hour), LFP chemistry
Power skids	6 skids, each 2.5 MW / 10 MWh (transformer + inverter + battery)
Data-center load	~38 MW facility (PUE ≈ 1.33)
Role	Peak-shaving + backup for the data center

Flow: Grid (138 kV) → main transformer → 34.5 kV bus → 6 feeders → power skids (each steps down to 690 V and ties a battery in via a bidirectional inverter) → data-center load.

2 • Topology

```
{
  "assets": [
    { "id": "SUBSTATION", "type": "substation", "label": "Grid / Substation (138 kV)", "x": 80, "y": 160 },
    { "id": "SKID-1", "type": "skid", "label": "Power Skid 1", "x": 300, "y": 40 },
    { "id": "SKID-2", "type": "skid", "label": "Power Skid 2", "x": 300, "y": 110 },
    { "id": "SKID-3", "type": "skid", "label": "Power Skid 3", "x": 300, "y": 180 },
    { "id": "SKID-4", "type": "skid", "label": "Power Skid 4", "x": 300, "y": 250 },
    { "id": "SKID-5", "type": "skid", "label": "Power Skid 5", "x": 300, "y": 320 },
    { "id": "SKID-6", "type": "skid", "label": "Power Skid 6", "x": 300, "y": 390 },
    { "id": "LOAD", "type": "load", "label": "Data Center Load", "x": 540, "y": 215 }
  ],
  "links": [
    { "from": "SUBSTATION", "to": "SKID-1" }, { "from": "SUBSTATION", "to": "SKID-2" },
    { "from": "SUBSTATION", "to": "SKID-3" }, { "from": "SUBSTATION", "to": "SKID-4" },
    { "from": "SUBSTATION", "to": "SKID-5" }, { "from": "SUBSTATION", "to": "SKID-6" },
    { "from": "SKID-1", "to": "LOAD" }, { "from": "SKID-2", "to": "LOAD" },
    { "from": "SKID-3", "to": "LOAD" }, { "from": "SKID-4", "to": "LOAD" },
    { "from": "SKID-5", "to": "LOAD" }, { "from": "SKID-6", "to": "LOAD" }
  ]
}
```

3 • Nameplate specifications

Asset	Rating
Grid interconnection	138 kV, 3-phase, 60 Hz
Main power transformer	138 / 34.5 kV, 50 MVA, on-load tap changer

Medium-voltage bus	34.5 kV
Skid transformer (×6)	34.5 kV / 690 V, 3 MVA
Inverter / PCS (×6)	2.5 MW bidirectional, 690 V AC, up to 1500 V DC
Battery container (×6)	10 MWh LFP, ~1330 V nominal DC
Data-center load	~38 MW facility

Battery internals (typical LFP): 3.2 V / 314 Ah cells (~1.0 kWh each); ~416 cells in series per string (~1331 V nominal); ~24 parallel strings per container ≈ 10 MWh.

4 • Telemetry schema

Every asset reports a `state` ∈ `NORMAL` | `WARNING` | `FAULT` | `OFFLINE` and an `alarms` array. Metric fields by asset type:

Substation / grid

field	unit	normal range
voltage_kV	kV	131–145 (138 ± 5%)
frequency_Hz	Hz	59.95–60.05
power_MW	MW	±50 · + = import, – = export
power_factor	–	0.95–1.00
main_tx_oil_temp_C	°C	40–75
main_tx_loading_pct	%	0–100

Skid → inverter (PCS)

field	unit	normal range
power_kW	kW	–2500...+2500 · + = charge, – = discharge
mode	enum	CHARGE / DISCHARGE / IDLE / FAULT
ac_voltage_V	V	~690
ac_current_A	A	0–2100
dc_voltage_V	V	1150–1500
efficiency_pct	%	96–99
igbt_temp_C	°C	30–65 (warn > 75)

Skid → battery

field	unit	normal range
soc_pct	%	10–95
soh_pct	%	80–100 (warn < 80)
dc_bus_V	V	1150–1500

current_A	A	0–1900
power_kW	kW	–2500...+2500
c_rate	C	0–0.25
cell_v_min / avg / max	V	2.80–3.65
cell_temp_min / avg / max	°C	15–40 (warn > 40)
cell_temp_delta_C	°C	0–8 (warn > 8)
insulation_MOhm	MΩ	> 1.0
strings_online	count	of 24
envelope.max_charge_kW	kW	0–2500
envelope.max_discharge_kW	kW	0–2500

The **operating envelope** is the max charge/discharge power the battery safely allows right now. It can be **derated** below nameplate (e.g., when a cell is hot). Show it in the panel — an operator needs to know *why* a skid can't deliver full power.

Skid → transformer · Data-center load

field	unit	normal range
transformer.temp_C	°C	40–90
transformer.loading_pct	%	0–100
load.power_MW	MW	0–40
load.it_load_MW	MW	0–30
load.pue	–	1.2–1.5

5 · Alarm catalog

code	severity	trigger	meaning
CELL_OV_WARN	warning	cell V > 3.55	approaching overvoltage
CELL_OV	critical	cell V > 3.65	overvoltage
CELL_UV_WARN	warning	cell V < 3.00	approaching undervoltage
CELL_UV	critical	cell V < 2.80	undervoltage
TEMP_HIGH	warning	cell temp > 40 °C	elevated temperature
TEMP_CRIT	critical	cell temp > 50 °C	over-temp — derate/stop
TEMP_DELTA	warning	max – min cell temp > 8 °C	thermal imbalance
SOC_LOW	warning	SoC < 10%	low state of charge
INSULATION_LOW	warning	IMD < 1.0 MΩ	insulation degraded
INSULATION_CRIT	critical	IMD < 0.5 MΩ	ground-fault risk
DC_OVERCURRENT	critical	current > rating	overcurrent

COMMS_LOST	critical	no telemetry from asset	asset offline
HVAC_FAULT	warning	cooling system fault	thermal management degraded
SOH_DEGRADED	warning	SoH < 80%	end-of-life approaching
GRID_FREQ	warning	freq outside 59.5–60.5 Hz	grid frequency excursion

6 • Live snapshot (one frame to render)

Scenario: the BESS is discharging to support the data-center load. Skid 2 has an elevated-temperature warning (and is derated); Skid 5 is offline (comms lost). Grid import $\approx$ facility load – BESS output.

```
{
  "stale": false,
  "assets": {
    "SUBSTATION": {
      "state": "NORMAL",
      "metrics": { "voltage_kV": 138.4, "frequency_Hz": 60.01, "power_MW": 28.5,
        "power_factor": 0.994, "main_tx_oil_temp_C": 58, "main_tx_loading_pct": 57 },
      "alarms": []
    },
    "SKID-1": {
      "state": "NORMAL",
      "pcs": { "state": "NORMAL", "power_kw": -1984, "mode": "DISCHARGE", "ac_voltage_V": 689,
        "ac_current_A": 1673, "dc_voltage_V": 1330, "efficiency_pct": 98.2,
        "igbt_temp_C": 48 },
      "battery": { "state": "NORMAL", "soc_pct": 61, "soh_pct": 97.3, "dc_bus_V": 1330,
        "current_A": 1519, "power_kw": -2020, "c_rate": 0.20,
        "cell_v_min": 3.181, "cell_v_avg": 3.198, "cell_v_max": 3.229,
        "cell_temp_min_C": 29.4, "cell_temp_avg_C": 31.6, "cell_temp_max_C": 33.8,
        "cell_temp_delta_C": 4.4, "insulation_MOhm": 3.2, "strings_online": 24,
        "envelope": { "max_charge_kw": 2500, "max_discharge_kw": 2500 } },
      "transformer": { "state": "NORMAL", "temp_C": 63, "loading_pct": 79 },
      "alarms": []
    },
    "SKID-2": {
      "state": "WARNING",
      "pcs": { "state": "NORMAL", "power_kw": -1480, "mode": "DISCHARGE", "ac_voltage_V": 690,
        "ac_current_A": 1247, "dc_voltage_V": 1333, "efficiency_pct": 98.1,
        "igbt_temp_C": 52 },
      "battery": { "state": "WARNING", "soc_pct": 58, "soh_pct": 96.1, "dc_bus_V": 1333,
        "current_A": 1132, "power_kw": -1509, "c_rate": 0.15,
        "cell_v_min": 3.176, "cell_v_avg": 3.205, "cell_v_max": 3.241,
        "cell_temp_min_C": 33.1, "cell_temp_avg_C": 37.8, "cell_temp_max_C": 41.2,
        "cell_temp_delta_C": 8.1, "insulation_MOhm": 2.7, "strings_online": 24,
        "envelope": { "max_charge_kw": 2500, "max_discharge_kw": 1500 } },
      "transformer": { "state": "NORMAL", "temp_C": 66, "loading_pct": 60 },
      "alarms": [
        { "code": "TEMP_HIGH", "severity": "warning", "message": "Battery module temperature
elevated (max 41.2 C)" },
        { "code": "TEMP_DELTA", "severity": "warning", "message": "Cell temperature spread 8.1 C –
discharge derated" }
      ],
    },
    "SKID-3": {
      "state": "NORMAL",
```

```

    "pcs": { "state": "NORMAL", "power_kw": -2060, "mode": "DISCHARGE", "dc_voltage_V": 1329,
    "igbt_temp_C": 49 },
    "battery": { "state": "NORMAL", "soc_pct": 60, "soh_pct": 97.6, "dc_bus_V": 1329,
    "current_A": 1578,
        "power_kw": -2097, "c_rate": 0.21, "cell_v_min": 3.178, "cell_v_avg": 3.195,
    "cell_v_max": 3.224,
        "cell_temp_min_C": 28.9, "cell_temp_avg_C": 31.1, "cell_temp_max_C": 33.2,
    "cell_temp_delta_C": 4.3,
        "insulation_MOhm": 3.4, "strings_online": 24,
        "envelope": { "max_charge_kw": 2500, "max_discharge_kw": 2500 } },
    "transformer": { "state": "NORMAL", "temp_C": 64, "loading_pct": 82 },
    "alarms": []
},
"SKID-4": {
    "state": "NORMAL",
    "pcs": { "state": "NORMAL", "power_kw": -1912, "mode": "DISCHARGE", "dc_voltage_V": 1331,
    "igbt_temp_C": 47 },
    "battery": { "state": "NORMAL", "soc_pct": 62, "soh_pct": 98.0, "dc_bus_V": 1331,
    "current_A": 1464,
        "power_kw": -1948, "c_rate": 0.19, "cell_v_min": 3.184, "cell_v_avg": 3.199,
    "cell_v_max": 3.226,
        "cell_temp_min_C": 27.8, "cell_temp_avg_C": 30.4, "cell_temp_max_C": 32.6,
    "cell_temp_delta_C": 4.8,
        "insulation_MOhm": 3.6, "strings_online": 24,
        "envelope": { "max_charge_kw": 2500, "max_discharge_kw": 2500 } },
    "transformer": { "state": "NORMAL", "temp_C": 61, "loading_pct": 76 },
    "alarms": []
},
"SKID-5": {
    "state": "OFFLINE",
    "pcs": null, "battery": null, "transformer": null,
    "alarms": [ { "code": "COMMS_LOST", "severity": "critical", "message": "No telemetry received
from skid" } ]
},
"SKID-6": {
    "state": "NORMAL",
    "pcs": { "state": "NORMAL", "power_kw": -1960, "mode": "DISCHARGE", "dc_voltage_V": 1332,
    "igbt_temp_C": 48 },
    "battery": { "state": "NORMAL", "soc_pct": 59, "soh_pct": 97.1, "dc_bus_V": 1332,
    "current_A": 1500,
        "power_kw": -1998, "c_rate": 0.20, "cell_v_min": 3.180, "cell_v_avg": 3.202,
    "cell_v_max": 3.231,
        "cell_temp_min_C": 29.0, "cell_temp_avg_C": 31.9, "cell_temp_max_C": 34.1,
    "cell_temp_delta_C": 5.1,
        "insulation_MOhm": 3.1, "strings_online": 24,
        "envelope": { "max_charge_kw": 2500, "max_discharge_kw": 2500 } },
    "transformer": { "state": "NORMAL", "temp_C": 62, "loading_pct": 78 },
    "alarms": []
},
"LOAD": {
    "state": "NORMAL",
    "metrics": { "power_MW": 37.9, "it_load_MW": 28.5, "pue": 1.33, "voltage_kV": 34.5 },
    "alarms": []
}
}
}

```

Power balance (sanity check): total skid discharge ≈ 9.4 MW; grid import 28.5 MW + BESS 9.4 MW ≈ 37.9 MW load. Small mismatches are normal (conversion losses, measurement).

7 · Simulating a live feed

Push updated frames periodically. To keep it believable:

- **Jitter analog values ± 0.3 – 0.5%** each frame (voltages, currents, temps, power).
- **Keep the balance:** `grid.power_MW  $\approx$  load.power_MW -  $\Sigma$ (skid discharge)` .
- **Occasionally change a state** — e.g., clear Skid 2's warning as it cools, or bring Skid 5 back online — so the UI's state transitions are visible.
- **Simulate a dropout** by setting `"stale": true` (or simply stopping the frames) so we can see how your UI flags a disconnected feed.
- **Fire an alarm burst** — e.g. 15+ alarms across the skids within a few seconds — so we can see your flood handling (Stage 1).